

Equalizer Lowpass FIR Filter Design via Complex Remez Algorithm

1 Complex Remez Algorithm

The complex Remez algorithm (also known as Lawson’s iteratively reweighted least-squares, IRLS) extends the classical Parks–McClellan algorithm to complex-valued FIR filters. At each iteration the algorithm solves a weighted least-squares problem

$$\min_{\mathbf{h}} \sum_k w_k(\ell) |\mathbf{a}(\omega_k)^\top \mathbf{h} - H_d(\omega_k)|^2,$$

and then updates the weights as $w_k(\ell+1) = w_k(\ell) |e_k(\ell)|$, where $e_k(\ell)$ is the error at frequency ω_k after iteration ℓ . This reweighting progressively concentrates the design effort on the worst-case frequency points, driving the solution toward the minimax (Chebyshev) optimum.

No explicit passband or stopband tolerance parameters are required; the algorithm converges to the equi-ripple solution determined solely by the frequency grid and the desired response $H_d(\omega)$.

1.1 Cosine Transition-Band Constraint

Without a transition-band constraint the IRLS solver leaves the gap between the passband and stopband entirely unconstrained, which can cause the magnitude to overshoot or roll off slowly in that region. A soft constraint is introduced by appending a transition-band grid $\mathcal{T}$ to the unified frequency grid and supplying a cosine-tapered desired magnitude $H_d^{(\text{tb})}(\omega)$ that interpolates smoothly between the passband-edge level and the stopband target. Specifically, for a normalised parameter $t \in [0, 1]$ sweeping from the passband edge to the stopband edge,

$$H_d^{(\text{tb})}(t) = A_{\text{stop}} + (A_{\text{pass}} - A_{\text{stop}}) \cdot \frac{1 + \cos(\pi t)}{2},$$

where $A_{\text{pass}} = \max_{k \in \mathcal{P}} |H_d(\omega_k)|$ is the passband-edge magnitude and $A_{\text{stop}} = \delta_s$ is the stopband target. This raised-cosine shape has zero derivative at both endpoints, giving a gentle rolloff at the passband edge and a smooth approach to the stopband level without abrupt kinks. The transition-band points enter the IRLS cost with the same band weight as the stopband ($B_k = w_{\text{sb}}$), so the solver tracks the ramp target directly rather than simply suppressing energy toward zero.

2 Simulation

Figure 1 shows the measured magnitude and phase response of the RX and Switched RX (SRX) paths, captured with a network analyser. Figure 2 overlays the channel response and its inverse equalizer response, both DC-normalised. The equalizer compensates for the channel magnitude roll-off across the passband. Figure 3 shows the desired equalizer response $H_d(\omega)$ after adding a linear-phase trend, which shifts the target to a causal FIR training target. Figures 4 and 5 show the trained FIR filter response. The filter is designed by the complex Remez algorithm to match the desired equalizer response $H_d(\omega)$.

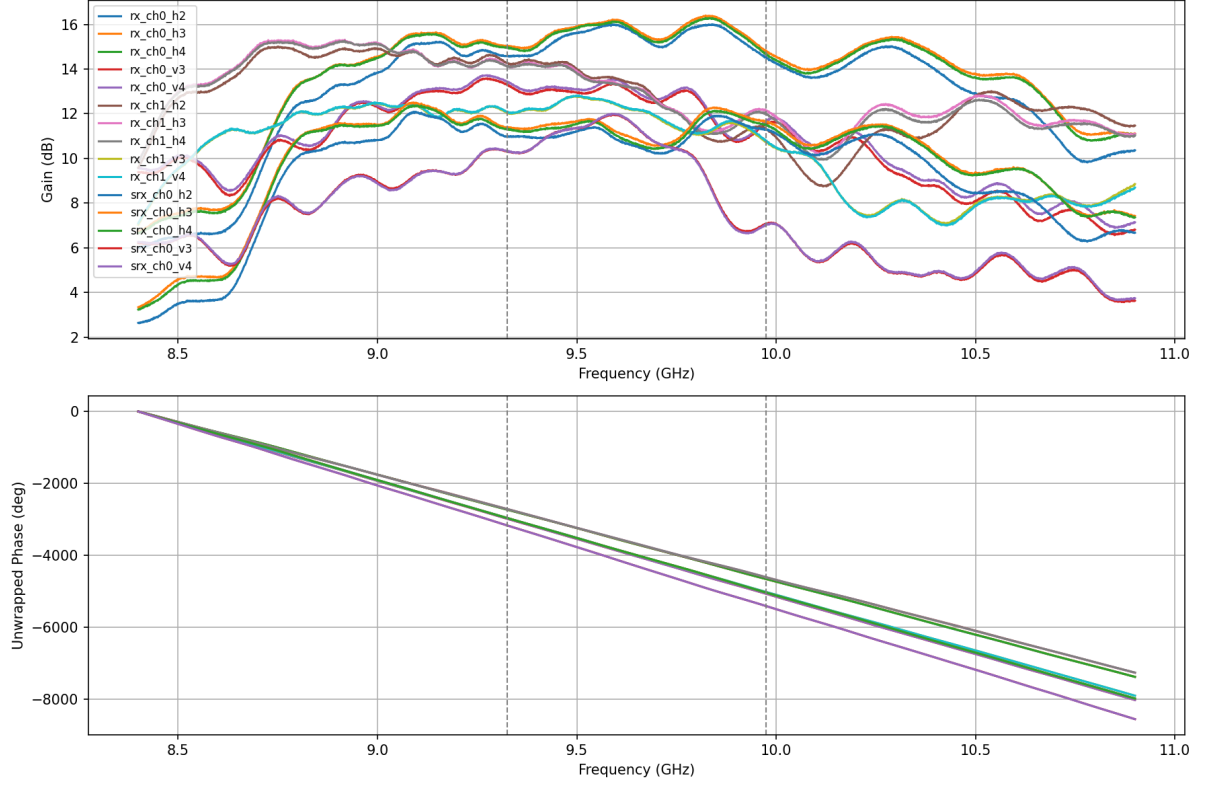


Figure 1: Measured RX and Switched RX (SRX) frequency responses from the network analyser. Top: magnitude (dB). Bottom: unwrapped phase (deg).

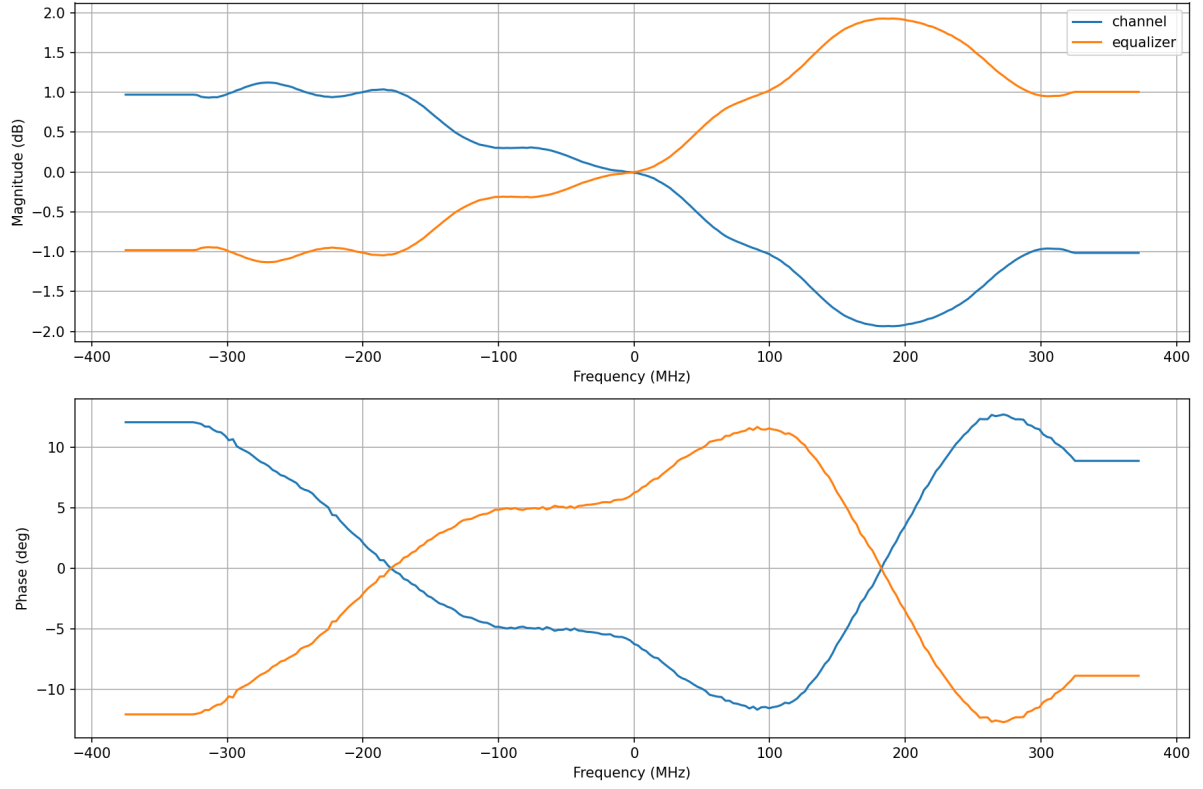


Figure 2: Channel and equalizer frequency responses overlaid. Top: magnitude (dB). Bottom: phase (deg).

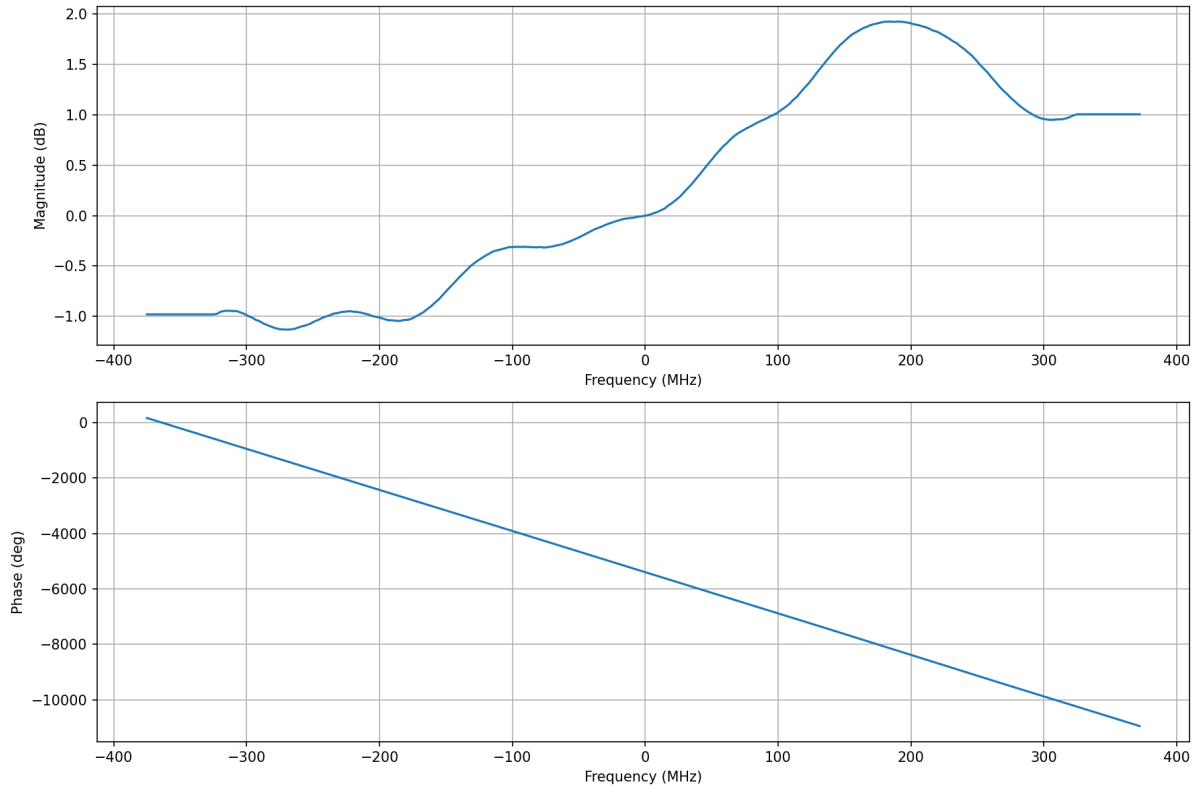


Figure 3: Desired equalizer frequency response $H_d(\omega)$ with linear-phase trend applied. Top: magnitude (dB). Bottom: phase (deg).

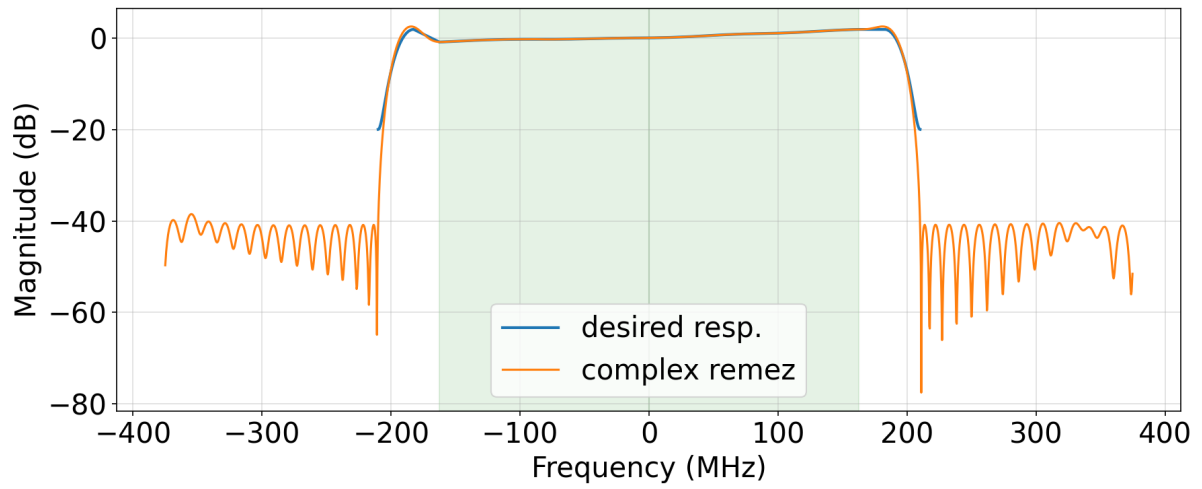


Figure 4: Full-band magnitude (top) and phase (bottom) of the trained equalizer FIR filter.

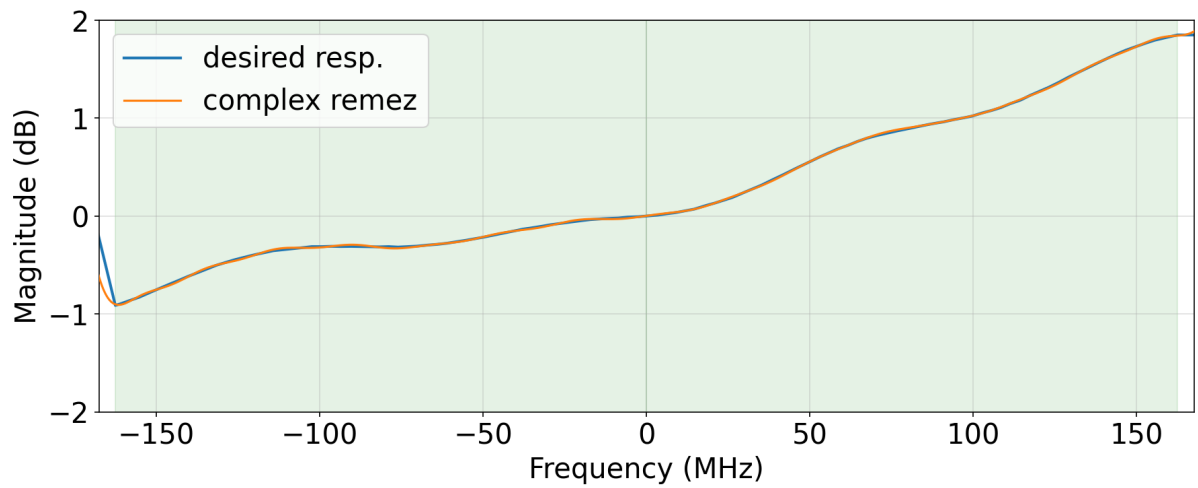


Figure 5: In-band magnitude response of the trained equalizer FIR filter, zoomed to the passband region.